Weighted Threshold Encryption with Silent Setup

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Abstract

We construct the first non-trivial weighted threshold encryption scheme with silent setup. The CRS and each public key consists of $O(W)$ group elements, where W is the maximum committee weight. Both the weight assignments and the threshold for decryption can be dynamically chosen at encryption time. Each partial decryption is a single $\mathbb{G}_1$ element computed *non-interactively* with one scalar multiplication, independent of the party's weight. The CPA ciphertext is two $\mathbb{G}_1$ elements, improving upon the shortest known *unweighted* scheme, and CCA security costs two additional field elements. We prove security in the generic bilinear group model.

1 Introduction

Threshold encryption [Des88, DF90] lets an encryptor produce a succinct ciphertext that any t out of n parties can jointly decrypt, while any coalition of fewer than t parties learns nothing. Traditionally, the parties first run an interactive distributed key generation (DKG) protocol to establish a shared public key, after which encryption is non-interactive and decryption consists of locally computed partial decryptions that anyone can aggregate.

A recently proposed notion of threshold encryption with *silent setup* [GKPW24] avoids interaction entirely during the setup phase. Each party samples a public key pair locally, independent of other parties. A message can be encrypted to an arbitrarily chosen committee with any threshold of choice given the committee's public keys. Decryption remains non-interactive, where each party computes partial decryptions that can be publicly aggregated. In deployments that motivate silent setup, typically with globally distributed participants such as distributed ledgers, trust is rarely uniform across all parties. In many proof-of-stake blockchains, the security of consensus rests on the assumption that the adversary controls less than a third of the *stake*, not validators. Any threshold system used by such a protocol, whether for encrypted mempools and MEV protection [CGPP24], timelock encryption [BF01], or VRFs [MRV99], must inherit this trust assumption.

The generic route to support weights is through *virtualization* where a party of weight w_i registers w_i keys and emulates w_i virtual parties in an unweighted scheme. This preserves the ciphertext size, but everything else in the protocol scales with a party's weight making it undesirable. Alternatively, when using Shamir secret sharing with threshold schemes that have linear aggregation, one can work in the *raised-hand setting* where the partial decryption algorithm knows which other parties will respond and can therefore internally aggregate its share of partial decryptions into a single element. In practice, this can lead to denial of service issues where a malicious party goes offline *intentionally* after raising its hand, thereby forcing the committee to redo the decryption.

While there exist constructions [BO22, GJM+23, GSW26] that improve upon naive virtualization and support batched decryption [ABD+25, FPRT26] of ciphertexts [CGPP24], all of them rely on an interactive setup, as we discuss below. In this work, we ask:

Can we realize efficient weighted threshold encryption without an interactive setup?[1]

[1] In fact, prior work [WW26, Section 1.1] by Waters and Wu lists efficient weighted threshold policies as an open problem.

In particular, we want to do better than naive virtualization, where partial decryption sizes grow with the weight of each party, and the public keys (including hints) scale as $O(w_i W)$, where w_i is the weight of the party and W is the maximum committee weight.

Our Contributions. We formalize weighted silent threshold encryption in Section 4 and construct the first *non-trivial* weighted threshold encryption scheme with silent setup:

- both the CRS and public keys are $O(W)$ group elements where W is the maximum committee weight
- the threshold for decryption and weight assignment can be chosen *dynamically* during encryption
- a partial decryption is one $\mathbb{G}_1$ element, computed *non-interactively* with one scalar multiplication
- the CPA ciphertext is two $\mathbb{G}_1$ elements – smaller than the shortest *unweighted* scheme [WW26]
- it can be upgraded to a CCA-secure version with two field elements via a sigma protocol

Prior work on STE. Garg, Kolonelos, Policharla, and Wang [GKPW24] built the first concretely efficient silent threshold encryption, proved secure in the generic bilinear group model. Both the public key and CRS grew linearly in the maximum committee size supported $O(N)$. Waters and Wu [WW26] reduced the ciphertext size and gave an alternate construction that was proved secure under a falsifiable q-type assumption in the plain model, but requires a quasilinear CRS $O(N \log N)$ to support dynamically chosen decryption thresholds. Champion, Wu, and Yamada [CWY26] gave a non-trivial construction of silent threshold encryption from lattices for sublinear thresholds.

The only work on weighted silent threshold encryption establishes feasibility using indistinguishability obfuscation [DJWW25]. In contrast, our work is a *concretely efficient* pairing-based construction.

Prior work on Weighted Threshold Encryption. Outside the silent setup model, there exist schemes that support weight-independent partial decryptions. Ferveo [BO22] runs a publicly verifiable DKG per epoch that outputs a public key of $O(w)$ group elements for a committee of total weight w. A partial decryption is one $\mathbb{G}_1$ element computed with one scalar multiplication, independent of the party's weight. We achieve similar characteristics *with* silent setup. Garg et al. [GJM+23] use CRT-based weighted *ramp* secret sharing to obtain $O(w_i)$-bit shares and one-element partial decryptions, but have a gap between the privacy and reconstruction thresholds, and work in the raised-hand setting. In [GSW26] both the secret keys and the communication are independent of the weights, and the scheme is proven secure under LWE, but decryption takes two rounds.

In the batched decryption setting motivated by encrypted mempools [CGPP24, CGPW25, AFP25, BFOQ25, BCF+25, BLT25, Pol26b, ADG+26, BNRT26], two recent works [ABD+25] and [FPRT26] support weight independent partial decryptions. BEAT++ [ABD+25] extends BEAT-MEV [BFOQ25] to the weighted setting: to decrypt a batch of B ciphertexts, each party publishes a single group element, regardless of its weight. Ciphertexts must, however, carry an index chosen at encryption time, and a batch can contain at most one ciphertext per index, inherited from prior work [BFOQ25]. More recently, [FPRT26] apply similar techniques to the index-free schemes Simple BTE and BTX [Pol26b, ADG+26] and the partial-fraction scheme of [BNRT26], so that ciphertexts are index independent and each party again publishes one $\mathbb{G}_1$ element per batch. In both cases the interactive setup fixes the committee and its weights, and the public key has $O(wB)$ group elements, where w is the weight of the committee.

2 Overview

We begin by introducing a useful polynomial identity at the heart of our construction. Throughout, W is the maximum total weight of a committee, $\boldsymbol{w} = (w_1, \ldots, w_n)$ are the weight assignments of the committee with $w = \sum_i w_i \leq W$, and $t \leq w$ is the decryption threshold.

Construction	Setup	$\lvert\mathsf{pp}\rvert$	$\lvert\mathsf{ct}\rvert$	$\lvert\mathsf{pd}_i\rvert$	Non-int. dec.
[BO22]	DKG	$O(w)$	g_1	g_1	✓
[GJM+23]	Dealer	$O(n) + O(w)$ bits	g	g	✗
[GSW26]	Dealer	$\mathsf{poly}(n, \lambda)$	$\mathsf{poly}(\lambda)$	$\mathsf{poly}(n, \lambda)$	✗
[ABD+25]	DKG	$O(wB)$	$2g_1 + g_2$	$g_1 + 2f$	✓
[FPRT26]	DKG	$O(wB)$	g_1	g_1	✓
[GKPW24] (virtualized)	Silent	$O(wW)$	$2g_1 + 7g_2$	$w_i g_2$	✓
[WW26] (virtualized)	Silent	$O(wW + W \log W)$	$g_1 + 2g_2 + f$	$w_i(g_1 + g_2)$	✓
This work	Silent	$O(nW)$	$2g_1$	g_1	✓

Table 1: Weighted threshold decryption for a committee of n parties with total weight w, where W is the maximum supported weight and B the batch size. $\lvert\mathsf{pp}\rvert$ counts all public material (CRS, public keys, and hints) in group elements unless stated otherwise. $\lvert\mathsf{ct}\rvert$ is the CPA ciphertext size ignoring the message. $\lvert\mathsf{pd}_i\rvert$ is the partial decryption size of a party with weight w_i, per ciphertext, or per batch for the batched schemes. $g_j = \lvert\mathbb{G}_j\rvert$ and $f = \lvert\mathbb{F}\rvert$ denote encoded lengths, and g the CRT scheme's group-element length. The last column indicates whether a partial decryption can be produced without knowing the responding set or additional rounds.

Polynomials with repeated roots. Fix distinct points $x_1, \ldots, x_n \in \mathbb{F}^*$, weights $w_i \in [1, W]$, and indeterminates $K_1, \ldots, K_n$, and define

$$A(X) = \sum_{i \in [n]} K_i \, (X - x_i)^{W - w_i}.$$

Now take any polynomial $B(X)$ of degree at most $W - 1$ and divide it by $(X - x_i)^{w_i}$ which gives

$$B(X) = (X - x_i)^{w_i} Q_i(X) + R_i(X)$$

where $\deg R_i < w_i$. The product $A(X) \cdot B(X)$ can then be written as

$$A(X) \cdot B(X) = \sum_{i \in [n]} K_i \, (X - x_i)^{W} Q_i(X) + \sum_{i \in [n]} K_i \, (X - x_i)^{W - w_i} R_i(X), \tag{1}$$

and in fact, this decomposition is unique (Lemma 2) when $\deg(R_i(X)) < w_i$.

Weights as Root Multiplicities. We now assign every party i a unique public label $x_i \in \mathbb{F}^*$. To prevent collisions this can be derived from the public key of each party. Each party also holds a secret key $\kappa_i \in \mathbb{F}^*$, and we set $K_i = \kappa_i$ and define the polynomial:

$$\mathsf{SK}(X) = \sum_{i \in [n]} \kappa_i \, (X - x_i)^{W - w_i}$$

which interpolates the committee's secret keys. Now observe that if we treat $B(X)$ as a selector polynomial indicating whether a party released their partial decryption we obtain:

$$\mathsf{SK}(X) \cdot B(X) = \underbrace{\sum_{i \in [n]} \kappa_i \, (X - x_i)^{W} Q_i(X)}_{Q(X)} + \underbrace{\sum_{i \in [n]} \kappa_i \, (X - x_i)^{W - w_i} R_i(X)}_{R(X)}.$$

Now observe that if we demand $\deg(B) \leq w - t$ and $B(X) \neq 0$, then by the pigeonhole principle:

$$\sum_{i : R_i(X) = 0} w_i \leq w - t \implies \sum_{i : R_i(X) \neq 0} w_i \geq t$$

since for each i, $R_i(X) = 0$ means that $B(X)$ has a root of multiplicity w_i at x_i, and a non-zero polynomial of degree at most $w - t$ has at most $w - t$ roots counted with multiplicity. This will allow us to design an encryption scheme such that:

- computing the contribution of the remainder component to the encapsulation key requires the help of committee members through partial decryptions
- we can simultaneously limit the total weight of parties *excluded* via the remainder to at most $w - t$ via a degree check on $B(X)$

Keys and hints. The setup samples $\tau \leftarrow\$ \mathbb{F}^*$ and publishes a powers-of-tau CRS $\{[\tau^j]_1, [\tau^j]_2\}_{j\in[0,2W-2]}$. Each party then samples $\mathsf{sk}, \kappa \leftarrow\$ \mathbb{F}^*$ and publishes:

$$\underbrace{\{[\kappa\tau^j]_1\}_{j\in[0,W-1]}}_{\mathsf{SK}(X)}, \qquad \underbrace{\{[\kappa\,(\tau - x)^j]_2\}_{j\in[W,2W-2]}}_{\text{quotient}}, \qquad \underbrace{\{[\mathsf{sk}\,\kappa\,(\tau - x)^j]_2\}_{j\in[0,W-1]}}_{\text{remainder}}.$$

together with $[\mathsf{sk}]_2$ (which will be used to verify partial decryptions). The additional sk in the remainder hints is what forces the aggregation to use the corresponding party's partial decryption. Given the committee and its weights, anyone can compute the commitment $\mathsf{com}_{\mathsf{pk}} = [\mathsf{SK}(\tau)]_1$ as a linear combination of the parties' $\mathbb{G}_1$ hints.

The relation. The encryptor witness encrypts [GKPW24, GHK+25, Pol26a] to the following relation, with highlighted witnesses:

$$\mathsf{com}_{\mathsf{pk}} \circ \boxed{B} = [1]_1 \circ \big(\boxed{R} + \boxed{Q}\big) \quad \text{(RootMul)}$$

$$[\tau^{2W-2-(w-t)}]_1 \circ \boxed{B} = [1]_1 \circ \boxed{\widehat{B}} \quad \text{(MaxDeg)}$$

$$[1]_1 \circ \boxed{B} = [\tau]_1 \circ \boxed{\bar{B}} + [1]_T \quad \text{(Eval)}$$

and produces a ciphertext of the form:

$$\mathsf{ct} = \big(r_1 \cdot \mathsf{com}_{\mathsf{pk}} + r_2[\tau^{2W-2-(w-t)}]_1 + r_3[1]_1,\ \ r_1[1]_1,\ \ r_2[1]_1,\ \ r_3[\tau]_1,\ \ \mathsf{msg} + r_3[1]_T\big),$$

The intuition behind each equation is as follows.

- RootMul: this is simply Eq. (1) evaluated at τ
- MaxDeg: since the CRS has no power of τ beyond τ^{2W-2} in $\mathbb{G}_2$, the witness $\widehat{B} = [\tau^{2W-2-(w-t)}B(\tau)]_2$ can only exist if $\deg B \le w - t$.
- Eval: $B(X) - 1 = X \cdot \bar{B}(X)$, i.e. $B(0) = 1$, so in particular $B \neq 0$.

Distributed decryption. An honest decryptor excludes the set of absent parties $[n] \setminus S$ by setting

$$B(X) = \prod_{i\in[n]\setminus S} \big(1 - X/x_i\big)^{w_i},$$

so that $B(0) = 1$, and since the weight of the responders in S is at least t, $\deg B = w - \sum_{i\in S} w_i \le w - t$. The decryptor then computes the witnesses

$$\boxed{B} = [B(\tau)]_2, \qquad \boxed{\widehat{B}} = [\tau^{2W-2-(w-t)}B(\tau)]_2, \qquad \boxed{\bar{B}} = [(B(\tau) - 1)/\tau]_2$$

from the CRS and

$$\boxed{Q} = \Big[\sum_{i\in[n]} \kappa_i(\tau - x_i)^W Q_i(\tau)\Big]_2$$

from the quotient hints, since every $(X - x_i)^W Q_i$ is a multiple of $(X - x_i)^W$ of degree $\leq 2W - 2$ (Lemma 1). While the polynomial identity guarantees that $R_i(X) = 0$ for all $i \in [n] \setminus S$, no party can compute

$$R = \left[\sum_{i \in S} \kappa_i (\tau - x_i)^{W - w_i} R_i(\tau)\right]_2$$

since the relevant hints carry the blinding sk_i. Instead, the set S that participates in decryption will release

$$\mathsf{pd}_i = \mathsf{sk}_i^{-1} \cdot \mathsf{ct}_2 = [\mathsf{sk}_i^{-1} r_1]_1$$

which will allow the aggregation to compute the *contribution* of $\{R_i\}_{i \in S}$ to the encapsulation key. Note that computing the partial decryption requires just one scalar multiplication, independent of its weight and does not require coordination between decryptors.

The aggregator can now compute

$$H_i = \left[\mathsf{sk}_i \, \kappa_i \, (\tau - x_i)^{W - w_i} R_i(\tau)\right]_2, \qquad \mathsf{pd}_i \circ H_i = \left[r_1 \, \kappa_i \, (\tau - x_i)^{W - w_i} R_i(\tau)\right]_T,$$

such that $\sum_{i \in S} \mathsf{pd}_i \circ H_i = \mathsf{ct}_2 \circ R$. The remaining witnesses B, Q, $\widehat{B}$, $\bar{B}$ can be used to complete the decryption and recover the message.

2.1 Reducing the ciphertext size

We now show how the ciphertext size can be reduced from four $\mathbb{G}_1$ elements to two $\mathbb{G}_1$ elements.

Absorb constraints into the CRS. As an instructive example, consider the KZG opening [KZG10] equation:

$$\mathsf{com} \circ [1]_2 = \pi \circ [\tau - x]_2 + [y]_1 \circ [1]_2.$$

With a CRS $\{[\tau^i]_1, [\tau^i]_2\}_{i \in [0,m]}$, the prover can commit to a polynomial of degree at most m. Now sample an independent $\alpha \leftarrow\!\!\$\, \mathbb{F}^*$, append $\left(\{[\alpha\tau^i]_1\}_{i \in [1,m']}, [\alpha^{-1}]_2\right)$ to the CRS, and modify the verification:

$$\mathsf{com} \circ [\alpha^{-1}]_2 = \pi \circ [\tau - x]_2 + [y]_1 \circ [1]_2.$$

In the GGM, com must now be a commitment to a polynomial of degree at most m' with constant term zero: the prover is forced to use $\{[\alpha\tau^i]_1\}_{i \in [1,m']}$ to construct the commitment, as otherwise the α^{-1} on the other side does not cancel.

Degree bound in the CRS. Replace the powers of τ used to commit the selector with

$$\{[\alpha\tau^{-j}]_2\}_{j \in [0, W-1]}, \qquad \{[\alpha^{-1}\tau^j]_1\}_{j \in [0, 2W-2]},$$

let each party publish $\{[\alpha^{-1}\kappa\tau^j]_1\}_{j \in [0,2W-2]}$ in place of $\{[\kappa\tau^j]_1\}_j$, and let preprocessing commit, for every threshold $t \in [1, w]$, to

$$\mathsf{com}_{\mathsf{pk},t} = \left[\alpha^{-1}\tau^{w-t}\,\mathsf{SK}(\tau)\right]_1.$$

The selector is now committed as $B = [\alpha\tau^{-(w-t)}B(\tau)]_2$, which requires $\deg B \leq w - t$ since only τ^{-j} for $j \leq W - 1$ is available, and no other $\mathbb{G}_2$ element of the CRS or of the hints carries a power of α.

Normalization from a hidden scalar. The Eval gadget only enforces $B(0) = 1$. Instead, sample $\gamma \leftarrow\$ \mathbb{F}^*$, shift the commitment,

$$\mathsf{com}_{\mathsf{pk},t} = \left[\alpha^{-1}\tau^{w-t}\left(\mathsf{SK}(\tau) + \gamma\right)\right]_1,$$

and change the relation to

$$\mathsf{com}_{\mathsf{pk},t} \circ \boxed{B} = [1]_1 \circ \left(\boxed{R} + \boxed{Q}\right) + [\gamma]_T.$$

To enable this, extend the CRS with

$$[\gamma]_T, \qquad \{[\gamma\alpha^{-1}\tau^j]_1\}_{j\in[0,W-1]}, \qquad \{[\gamma\tau^j]_2\}_{j\in[1,W-1]}.$$

The first family lets preprocessing add $[\gamma\alpha^{-1}\tau^{w-t}]_1$ to the commitment. For an honest prover the shift adds $\gamma B(X)$ to the product on the left,

$$\left(\mathsf{SK}(X) + \gamma\right) B(X) = \mathsf{SK}(X)B(X) + \gamma + \gamma\big(B(X) - 1\big),$$

and since $B(0) = 1$, the polynomial $B(X) - 1 = \sum_{j=1}^{\deg B} b_j X^j$ has no constant term, so the correction is a combination of the new $\mathbb{G}_2$ elements and can be folded into the public witness:

$$\boxed{Q} = \Big[\sum_{i\in[n]} \kappa_i(\tau - x_i)^W Q_i(\tau) + \gamma\big(B(\tau) - 1\big)\Big]_2 = \Big[\sum_{i\in[n]} \kappa_i(\tau - x_i)^W Q_i(\tau)\Big]_2 + \sum_{j=1}^{\deg B} b_j \cdot [\gamma\tau^j]_2.$$

For soundness, treat τ, α, γ and the parties' scalars as formal variables and compare the terms of the relation that are linear in γ. On the left, $\mathsf{com}_{\mathsf{pk},t} \circ \boxed{B}$ contributes $\gamma B(X)$, where B is a Laurent polynomial with exponents at most $w - t$ because $\boxed{B}$ was built from $\{[\alpha\tau^{-j}]_2\}_{j\in[0,W-1]}$. On the right, the only $\mathbb{G}_2$ elements involving γ are $\{[\gamma\tau^j]_2\}_{j\in[1,W-1]}$, and $[\gamma]_T$, so

$$\gamma\, B(X) = \gamma \sum_{j=1}^{W-1} c_j X^j + \gamma \qquad \Longrightarrow \qquad B(X) = 1 + \sum_{j=1}^{W-1} c_j X^j.$$

Hence B is an ordinary polynomial, of degree at most $w - t$ by the exponent bound, with $B(0) = 1$.

The resulting scheme. The relation is now a single equation, and a witness encryption to it samples $r \leftarrow\$ \mathbb{F}^*$ and outputs

$$\mathsf{ct} = (\mathsf{ct}_1, \mathsf{ct}_2, \mathsf{ct}_3) = \big(r \cdot \mathsf{com}_{\mathsf{pk},t},\ [r]_1,\ \mathsf{msg} + r \cdot [\gamma]_T\big).$$

Decryption is unchanged: party i responds with $\mathsf{pd}_i = \mathsf{sk}_i^{-1} \cdot \mathsf{ct}_2$, the aggregator forms $\boxed{B}$, $\boxed{Q}$, and the H_i and outputs

$$\mathsf{msg} = \mathsf{ct}_3 - \mathsf{ct}_1 \circ \boxed{B} + \mathsf{ct}_2 \circ \boxed{Q} + \sum_{i\in S} \mathsf{pd}_i \circ H_i.$$

The security argument is the one given above, with α and γ taking over the roles of MaxDeg and Eval respectively. The above scheme can be upgraded to CCA security by attaching a simulation-extractable NIZK proof of knowledge of the discrete logarithm of $\mathsf{ct}_2 = [r]_1$ incurring two field elements via a Sigma protocol.

We provide the full construction in Fig. 1, and a security proof in Section 5.2.

3 Preliminaries

Bilinear groups and polynomial notation. We use asymmetric groups $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ of prime order p, with a non-degenerate bilinear map $\circ : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ and no maps between the source groups. In additive exponent notation, $[a]_1 \circ [b]_2 = [ab]_T$. The generators $[1]_1, [1]_2$ and their pairing are public. The field is $\mathbb{F} = \mathbb{F}_p$, with p exponentially large in the security parameter and $p > 2W$. We do not require $W \mid p - 1$. For a monic polynomial D, $P \bmod D$ is the unique remainder of degree less than $\deg D$. The notation $\mathbb{F}[X]_{\le d}$ denotes polynomials of degree at most d, and is $\{0\}$ for $d < 0$.

Generic bilinear group model. Our security proof is based on the Generic Group Model [Sho97, Mau05]. A generic adversary does not have concrete representations of the elements of the group and can only use generic group operations.

In particular, we follow Maurer's GGM [Mau05], which is extended to bilinear groups by [BBG05]. The adversary in GGM makes oracle queries for each generic group operation she wishes to perform and receives a handle for the resulting group element, instead of the actual element itself. We call the party that answers the queries the challenger. The challenger keeps three lists L_1, L_2, L_T of all group elements resulting from the queries of the adversary together with their corresponding handles.

A standard GGM technique in security proofs is symbolic equivalence. We call symbolic experiment (and symbolic group representation, respectively) the model where the challenger is storing polynomials instead of group elements and performs polynomial operations instead of group operations. The formal variables of the polynomials are the initial elements that the adversary received.

Definition 1. *Let L_1, L_2, L_T be lists of polynomials in $\mathbb{F}[Y_1, \ldots, Y_k]$ and let $\mathcal{C}(L) = \{g_1 \cdot g_2 : g_1 \in L_1,\, g_2 \in L_2\} \cup L_T$ be their completion. A polynomial f is* dependent *on $L = (L_1, L_2, L_T)$ if $f \in \operatorname{span}_{\mathbb{F}} \mathcal{C}(L)$, and* independent *otherwise.*

Theorem 1 (Master theorem [BBG05, Boy08])**.** *Let L_1, L_2, L_T be lists of ν polynomials in total, of degree at most d, and let f be a polynomial of degree at most d that is independent of (L_1, L_2, L_T). For every generic adversary $\mathcal{A}$ making at most q oracle queries,*

$$\left|\Pr\left[\mathcal{A}\big(\mathcal{H},\, h_T[f(\boldsymbol{y})]\big) = 1\right] - \Pr\left[\mathcal{A}\big(\mathcal{H},\, h_T[\rho]\big) = 1\right]\right| \leq \frac{(q+\nu+2)^2\, d}{2p},$$

where $\mathcal{H} = \big(h_1[L_1(\boldsymbol{y})], h_2[L_2(\boldsymbol{y})], h_T[L_T(\boldsymbol{y})]\big)$ denotes the handles to the lists, and the probability is over $\boldsymbol{y} \leftarrow \mathbb{F}^k$, $\rho \leftarrow \mathbb{F}$, and the handle assignment.

Non-interactive zero-knowledge proofs. Let $\mathcal{L}$ be an NP language and $\mathcal{R}$ its corresponding NP relation. A simulation-extractable NIZK for $\mathcal{R}$ consists of the following algorithms:

- $\mathsf{Setup}(1^\lambda) \to (\mathsf{crs}, \mathsf{td})$: takes a security parameter and outputs a common reference string crs and trapdoor $\mathsf{td} = (\mathsf{td}_{\mathsf{sim}}, \mathsf{td}_{\mathsf{ext}})$.
- $\mathsf{Prove}(\mathsf{crs}, x, w) \to \pi$: takes $(x, w) \in \mathcal{R}$ and outputs a proof π for $x \in \mathcal{L}$.
- $\mathsf{Verify}(\mathsf{crs}, x, \pi) \to b$: outputs a bit indicating whether verification succeeds.
- $\mathsf{SimProve}(\mathsf{crs}, \mathsf{td}_{\mathsf{sim}}, x) \to \pi$: outputs a simulated proof for an arbitrary statement x.

We drop the crs argument wherever it is implicit. When specifying a relation, we write $\{w \mid \varphi_1 \wedge \cdots \wedge \varphi_m\}$, where w is the witness and each φ_i is a constraint. A proof *bound to* a tag tg is a proof in the tag-based NIZK whose statement is (x, tg). In particular a proof for one tag does not verify under another.

Definition 2 (Straight-line simulation-extractable NIZK)**.** *A simulation-extractable NIZK satisfies the following properties.*

- Perfect completeness. *For every $(x, w) \in \mathcal{R}$,*

$$\Pr\left[\begin{array}{c}(\mathsf{crs}, \mathsf{td}) \leftarrow \mathsf{Setup}(1^\lambda) \\ \pi \leftarrow \mathsf{Prove}(\mathsf{crs}, x, w)\end{array} : \mathsf{Verify}(x, \pi) = 1\right] = 1.$$

- Proof of knowledge. *For every PPT adversary $\mathcal{A}$, there is a PPT extractor* Ext *such that*

$$\Pr\left[\begin{array}{c}(\mathsf{crs}, \mathsf{td}) \leftarrow \mathsf{Setup}(1^\lambda) \\ (x, \pi) \leftarrow \mathcal{A}(\mathsf{crs}) \\ w \leftarrow \mathsf{Ext}(\mathsf{crs}, \mathsf{td}_{\mathsf{ext}}, x, \pi; \mathsf{aux})\end{array} : \begin{array}{c}\mathsf{Verify}(x, \pi) = 1 \\ (x, w) \notin \mathcal{R}\end{array}\right] \leq \mathsf{negl}(\lambda).$$

- Simulated-proof correctness. *For every statement* x*,*

$$\Pr\left[\begin{matrix}(\mathsf{crs},\mathsf{td})\leftarrow\mathsf{Setup}(1^\lambda)\\ \pi\leftarrow\mathsf{SimProve}(\mathsf{crs},\mathsf{td}_{\mathsf{sim}},x)\end{matrix}:\mathsf{Verify}(x,\pi)=1\right]=1.$$

- Computational zero knowledge. *There exists an algorithm* SimProve *such that, for every PPT adversary* $\mathcal{A}$*, define*

$$p_{\mathsf{real}}=\Pr\left[\begin{matrix}(\mathsf{crs},\mathsf{td})\leftarrow\mathsf{Setup}(1^\lambda)\\ (x,w)\leftarrow\mathcal{A}(\mathsf{crs})\\ \pi\leftarrow\mathsf{Prove}(\mathsf{crs},x,w)\end{matrix}:\begin{matrix}(x,w)\in\mathcal{R}\\ \mathcal{A}(\pi)=1\end{matrix}\right],$$

$$p_{\mathsf{sim}}=\Pr\left[\begin{matrix}(\mathsf{crs},\mathsf{td})\leftarrow\mathsf{Setup}(1^\lambda)\\ (x,w)\leftarrow\mathcal{A}(\mathsf{crs})\\ \pi\leftarrow\mathsf{SimProve}(\mathsf{crs},\mathsf{td}_{\mathsf{sim}},x)\end{matrix}:\begin{matrix}(x,w)\in\mathcal{R}\\ \mathcal{A}(\pi)=1\end{matrix}\right].$$

  Computational zero knowledge requires

$$|p_{\mathsf{real}}-p_{\mathsf{sim}}|\le\mathsf{negl}(\lambda).$$

  We require the same indistinguishability for polynomially many valid statement–witness pairs chosen adaptively under one crs*.*

- Simulation extractability. *For every PPT adversary* $\mathcal{A}$*, there exists a PPT extractor* Ext*. On any statement* x*, the simulation oracle computes*

$$\pi\leftarrow\mathsf{SimProve}(\mathsf{crs},\mathsf{td}_{\mathsf{sim}},x),$$

  adds (x,π) *to* Q*, and returns* π*. Then*

$$\Pr\left[\begin{matrix}(\mathsf{crs},\mathsf{td})\leftarrow\mathsf{Setup}(1^\lambda),\quad Q\leftarrow\emptyset\\ (x,\pi)\leftarrow\mathcal{A}^{\mathcal{O}_{\mathsf{sim}}}(\mathsf{crs})\\ w\leftarrow\mathsf{Ext}(\mathsf{crs},\mathsf{td}_{\mathsf{ext}},x,\pi;\mathsf{aux})\end{matrix}:\begin{matrix}\mathsf{Verify}(x,\pi)=1\\ (x,w)\notin\mathcal{R}\\ (x,\pi)\notin Q\end{matrix}\right]\le\mathsf{negl}(\lambda).$$

The extractor is straight line: on input a valid proof and aux*, it extracts a witness with overwhelming probability without rewinding the prover. In the standard CRS model,* aux *is empty. In the random-oracle model, it may contain the adversary's random-oracle queries. The* crs *may then be empty.*

4 Definitions

We extend the definition of silent threshold encryption [GKPW24, WW26] to weighted committees. A committee consists of n parties with public weights $\boldsymbol{w}=(w_1,\ldots,w_n)$ and total weight $w=\sum_{i\in[n]}w_i\le W$, where W is a bound fixed at setup. A set $S\subseteq[n]$ is authorized for threshold t if $\sum_{i\in S}w_i\ge t$.

Definition 3 (Weighted Silent Threshold Encryption)**.** *A weighted silent threshold encryption scheme is a tuple of algorithms* (Setup, KeyGen, Preprocess, Enc, PartDec, Verify, Aggregate) *with the following syntax.*

- $\mathsf{Setup}(1^\lambda,W)\to\mathsf{crs}$*: on input the security parameter and a bound* W *on the total committee weight, output a common reference string.*

- $\mathsf{KeyGen}(\mathsf{crs})\to(\mathsf{pk},\mathsf{sk})$*: on input the* crs*, output a key pair. The party may later be assigned any weight* $w_i\le W$*.*

- $\mathsf{Preprocess}(\mathsf{crs}, (\mathsf{pk}_1, \ldots, \mathsf{pk}_n), \boldsymbol{w}) \to (\mathsf{ek}, \mathsf{dk})$*: a deterministic algorithm that, on input the public keys of a committee and its weights, outputs an encryption key* ek *and a decryption key* dk.
- $\mathsf{Enc}(\mathsf{ek}, \mathsf{msg}, t) \to \mathsf{ct}$*: on input an encryption key, a message, and a threshold $t \leq w$, output a ciphertext.*
- $\mathsf{PartDec}(\mathsf{ek}, \mathsf{sk}, \mathsf{ct}) \to \mathsf{pd}$*: on input an encryption key, a secret key, and a ciphertext, output a partial decryption, or $\perp$ if the ciphertext is malformed.*
- $\mathsf{Verify}(\mathsf{ct}, \mathsf{pd}, \mathsf{pk}) \to 0/1$*: on input a ciphertext, a partial decryption, and a public key, output 1 iff the partial decryption is valid.*
- $\mathsf{Aggregate}(\mathsf{dk}, \mathsf{ct}, \{\mathsf{pd}_i\}_{i \in S}) \to \mathsf{msg}$*: on input a decryption key, a ciphertext, and partial decryptions from a set S, output a message.*

A secure weighted silent threshold encryption scheme must satisfy correctness and CCA security.

Definition 4 (Correctness)**.** *A weighted STE scheme is correct if for all* crs *in the support of* $\mathsf{Setup}(1^\lambda, W)$*, all n, all weight assignments $\boldsymbol{w}$ with $\sum_i w_i \leq W$, all $(\mathsf{pk}_i, \mathsf{sk}_i)$ in the support of* $\mathsf{KeyGen}(\mathsf{crs})$*, all messages* msg*, all thresholds $t \leq w$, and all* ct *in the support of* $\mathsf{Enc}(\mathsf{ek}, \mathsf{msg}, t)$*, the following hold, where* $(\mathsf{ek}, \mathsf{dk}) \leftarrow \mathsf{Preprocess}(\mathsf{crs}, (\mathsf{pk}_i)_{i \in [n]}, \boldsymbol{w})$.

- Partial decryption correctness: *for all $i \in [n]$,* $\mathsf{Verify}(\mathsf{ct}, \mathsf{PartDec}(\mathsf{ek}, \mathsf{sk}_i, \mathsf{ct}), \mathsf{pk}_i) = 1$.
- Aggregation correctness: *for all authorized $S \subseteq [n]$, i.e. $\sum_{i \in S} w_i \geq t$, and all partial decryptions $\{\mathsf{pd}_i\}_{i \in S}$ with* $\mathsf{Verify}(\mathsf{ct}, \mathsf{pd}_i, \mathsf{pk}_i) = 1$ *for all $i \in S$, we have* $\mathsf{Aggregate}(\mathsf{dk}, \mathsf{ct}, \{\mathsf{pd}_i\}_{i \in S}) = \mathsf{msg}$.

Note that aggregation correctness quantifies over *all* partial decryptions that verify, not only honestly generated ones, so a corrupt party cannot disrupt decryption by an authorized set.

Definition 5 (CCA Security)**.** *For an adversary $\mathcal{A}$ consider the following game.*

1. *$\mathcal{A}$ outputs the number of parties n, a weight assignment $\boldsymbol{w}$ with $w = \sum_i w_i \leq W$, and a set* $\mathsf{Cor} \subseteq [n]$ *of corrupt parties.*
2. *The challenger samples* $\mathsf{crs} \leftarrow \mathsf{Setup}(1^\lambda, W)$ *and, for $i \notin \mathsf{Cor}$,* $(\mathsf{pk}_i, \mathsf{sk}_i) \leftarrow \mathsf{KeyGen}(\mathsf{crs})$*, and gives* crs *and* $\{\mathsf{pk}_i\}_{i \notin \mathsf{Cor}}$ *to $\mathcal{A}$.*
3. *$\mathcal{A}$ specifies the key-generation randomness ρ_i for each $i \in \mathsf{Cor}$, and the challenger computes* $(\mathsf{pk}_i, \mathsf{sk}_i)$ *by running* $\mathsf{KeyGen}(\mathsf{crs})$ *with randomness ρ_i. It then computes* $(\mathsf{ek}, \mathsf{dk}) = \mathsf{Preprocess}(\mathsf{crs}, (\mathsf{pk}_i)_{i \in [n]}, \boldsymbol{w})$ *and gives both keys to $\mathcal{A}$.*
4. *$\mathcal{A}$ may make partial decryption queries (i, ct) for honest parties $i \notin \mathsf{Cor}$, and the challenger responds with* $\mathsf{PartDec}(\mathsf{ek}, \mathsf{sk}_i, \mathsf{ct})$.
5. *$\mathcal{A}$ outputs a threshold $t \leq w$ with $\sum_{i \in \mathsf{Cor}} w_i < t$ and messages* $\mathsf{msg}_0, \mathsf{msg}_1$*. The challenger samples a bit $b \leftarrow \{0, 1\}$ and returns the challenge* $\mathsf{ct}^* \leftarrow \mathsf{Enc}(\mathsf{ek}, \mathsf{msg}_b, t)$.
6. *$\mathcal{A}$ may continue to make partial decryption queries (i, ct) with $i \notin \mathsf{Cor}$. Queries with $\mathsf{ct} = \mathsf{ct}^*$ are answered as long as the total weight of* Cor *and of the honest parties queried on ct^* stays below t.*
7. *$\mathcal{A}$ outputs a bit b'.*

A weighted STE scheme is CCA-secure *if for every $W = \mathsf{poly}(\lambda)$ and every PPT $\mathcal{A}$,*

$$\left|\Pr[b' = 1 \mid b = 0] - \Pr[b' = 1 \mid b = 1]\right| = \mathsf{negl}(\lambda).$$

Definition 6 (Succinctness)**.** *A non-trivial weighted silent threshold encryption scheme satisfies the following efficiency requirements:* ct *is independent of n, $\boldsymbol{w}$, and W up to the description of $(t, \boldsymbol{w})$, and the size of* pd *and the running time of* PartDec *are constant, in particular independent of the party's weight.*

5 Construction

We now give the full construction in Fig. 1 corresponding to the two-element scheme discussed in Section 2.1. We then prove correctness in Section 5.1, and prove security in Section 5.2.

Weighted Silent Threshold Encryption

Setup. On input $(1^\lambda, W)$, sample $\alpha, \tau, \gamma \leftarrow\$ \mathbb{F}^*$ and publish

$$\{[\alpha\tau^{-j}]_2\}_{j\in[0,W-1]}, \quad \{[\alpha^{-1}\tau^j]_1\}_{j\in[0,2W-2]}, \quad \{[\tau^j]_2\}_{j\in[0,2W-2]},$$
$$[\gamma]_T, \quad \{[\gamma\alpha^{-1}\tau^j]_1\}_{j\in[0,W-1]}, \quad \{[\gamma\tau^j]_2\}_{j\in[1,W-1]}.$$

KeyGen. On input crs, sample a label $x \leftarrow\$ \mathbb{F}^*$, a key $\kappa \leftarrow\$ \mathbb{F}^*$, and a blinding $\mathsf{sk} \leftarrow\$ \mathbb{F}^*$. Publish

$$\mathsf{pk} = \Big(x,\ [\mathsf{sk}]_2,\ \{[\alpha^{-1}\kappa\tau^j]_1\}_{j\in[0,2W-2]},\ \{[\mathsf{sk}\,\kappa(\tau-x)^j]_2\}_{j\in[0,W-1]},\ \{[\kappa(\tau-x)^j]_2\}_{j\in[W,2W-2]}\Big)$$

and retain sk.

Preprocess. On input $(\mathsf{crs}, (\mathsf{pk}_i)_{i\in[n]}, \boldsymbol{w})$, reject unless $w_i > 0$ and $1 \le w = \sum_i w_i \le W$. Define

$$\mathsf{SK}(X) = \sum_{i\in[n]} \kappa_i\,(X - x_i)^{W-w_i}.$$

Using the $\mathbb{G}_1$ hints compute

$$\Big\{\mathsf{com}_{\mathsf{pk},t} = \big[\alpha^{-1}\tau^{w-t}\,(\mathsf{SK}(\tau)+\gamma)\big]_1\Big\}_{t\in[1,w]}$$

and output

$$\mathsf{ek} = (\boldsymbol{w}, \{\mathsf{com}_{\mathsf{pk},t}\}_{t\in[1,w]}), \qquad \mathsf{dk} = (\mathsf{ek}, (\mathsf{pk}_i)_{i\in[n]}).$$

Encryption. On input $(\mathsf{ek}, \mathsf{msg}, t)$ with $1 \le t \le w$, witness encrypt msg to the relation

$$\mathsf{com}_{\mathsf{pk},t} \circ B = [1]_1 \circ (R + Q) + [\gamma]_T :$$

sample $r \leftarrow\$ \mathbb{F}^*$, compute

$$(\mathsf{ct}_1, \mathsf{ct}_2, \mathsf{ct}_3) = \big(r\cdot\mathsf{com}_{\mathsf{pk},t},\ [r]_1,\ \mathsf{msg} + r\cdot[\gamma]_T\big)$$

and a proof of knowledge

$$\pi = \{\, r \mid \mathsf{ct}_2 = r\cdot[1]_1 \,\}$$

bound to $(t, \mathsf{ct}_1, \mathsf{ct}_3)$, and output $\mathsf{ct} = (t, \mathsf{ct}_1, \mathsf{ct}_2, \mathsf{ct}_3, \pi)$.

Partial Decryption. On input $(\mathsf{ek}, \mathsf{sk}, \mathsf{ct})$, output $\perp$ if π does not verify or $t \notin [1, w]$, and $\mathsf{pd} = \mathsf{sk}^{-1}\cdot\mathsf{ct}_2$ otherwise.

Verify. On input $(\mathsf{ct}, \mathsf{pd}_i, \mathsf{pk}_i)$, accept iff $\mathsf{pd}_i \circ [\mathsf{sk}_i]_2 = \mathsf{ct}_2 \circ [1]_2$.

Aggregate. On input $(\mathsf{dk}, \mathsf{ct}, \{\mathsf{pd}_i\}_{i\in S})$, reject unless π verifies, partial decryptions verify, and $\sum_{i\in S} w_i \ge t$. Define

$$B(X) = \prod_{i\in[n]\setminus S} (1 - X/x_i)^{w_i},$$

so that $B(0) = 1$, $\deg B = w - \sum_{i \in S} w_i \le w - t$, and B has a root of multiplicity w_i at x_i for every party $i \notin S$. For $i \in [n]$ let

$$R_i(X) = B(X) \bmod (X - x_i)^{w_i}, \qquad Q_i(X) = \frac{B(X) - R_i(X)}{(X - x_i)^{w_i}},$$

so that $\deg R_i < w_i$, $R_i = 0$ for $i \notin S$, and $(X - x_i)^{W - w_i} B = (X - x_i)^W Q_i + (X - x_i)^{W - w_i} R_i$. Compute

$$\begin{aligned} B &= \big[\alpha \tau^{-(w-t)} B(\tau)\big]_2, \\ Q &= \Big[\sum_{i \in [n]} \kappa_i (\tau - x_i)^W Q_i(\tau) + \gamma\big(B(\tau) - 1\big)\Big]_2, \\ H_i &= \big[\mathsf{sk}_i \, \kappa_i \, (\tau - x_i)^{W - w_i} R_i(\tau)\big]_2 \qquad (i \in S), \end{aligned}$$

where Q uses the unblinded hints, since $(X - x_i)^W Q_i$ is a combination of $(X - x_i)^j$ for $j \in [W, 2W - 2]$, and H_i the blinded ones, since $(X - x_i)^{W - w_i} R_i$ is a combination of $(X - x_i)^j$ for $j \in [W - w_i, W - 1]$. Output

$$\mathsf{msg} = \mathsf{ct}_3 - \mathsf{ct}_1 \circ B + \mathsf{ct}_2 \circ Q + \sum_{i \in S} \mathsf{pd}_i \circ H_i.$$

Figure 1: Our weighted threshold encryption scheme with silent setup.

5.1 Correctness

All hints of a party are powers of $\tau - x$ rather than powers of τ, and the aggregator works with B written in powers of $X - x_i$. This is a change of basis, not a restriction: every polynomial has a Taylor expansion around every point, over any field.

Lemma 1. *For every $x^* \in \mathbb{F}$ and $f \in \mathbb{F}[X]_{\le d}$ there exist unique $h_0, \ldots, h_d \in \mathbb{F}$ with*

$$f(X) = \sum_{k=0}^{d} h_k \, (X - x^*)^k,$$

and

$$\sum_{k < w} h_k (X - x^*)^k = f \bmod (X - x^*)^w$$

for every $w \ge 0$.

Proof. The polynomial $g(Y) := f(Y + x^*)$ has degree at most d, and writing $g(Y) = \sum_{k=0}^{d} h_k Y^k$ and substituting $Y = X - x^*$ gives the first identity. For uniqueness, the polynomials $(X - x^*)^0, \ldots, (X - x^*)^d$ have distinct degrees, so they are linearly independent and form a basis of $\mathbb{F}[X]_{\le d}$.

For the second identity, split the sum as

$$f(X) = \sum_{k < w} h_k (X - x^*)^k + (X - x^*)^w \sum_{k \ge w} h_k (X - x^*)^{k - w},$$

where the first sum has degree less than w and the second term is a multiple of $(X - x^*)^w$. By uniqueness of the remainder, the first sum equals $f \bmod (X - x^*)^w$. □

Lemma 2 (Multiplicity decomposition)**.** *Let $x_1, \ldots, x_n \in \mathbb{F}$, let $1 \le w_i \le W$, let $K_1, \ldots, K_n$ be indeterminates, and define*

$$A(X) = \sum_{i \in [n]} K_i \, (X - x_i)^{W - w_i} \in \mathbb{F}[K_1, \ldots, K_n][X].$$

For every $B \in \mathbb{F}[X]$ with $\deg B \le W-1$ there are unique $R_1,\dots,R_n,Q_1,\dots,Q_n \in \mathbb{F}[X]$ with $\deg R_i < w_i$ such that

$$A(X)\,B(X) = \sum_{i\in[n]} K_i\,(X-x_i)^{W-w_i} R_i(X) + \sum_{i\in[n]} K_i\,(X-x_i)^{W} Q_i(X), \tag{2}$$

namely

$$R_i(X) = B(X) \bmod (X-x_i)^{w_i}, \qquad Q_i(X) = \frac{B(X)-R_i(X)}{(X-x_i)^{w_i}}.$$

In particular $\deg Q_i \le W-2$, and $R_i = 0$ if and only if $(X-x_i)^{w_i} \mid B$.

Proof. Both sides of (2) are linear in the indeterminates K_i, so the identity holds if and only if for every i

$$(X-x_i)^{W-w_i} B(X) = (X-x_i)^{W-w_i} R_i(X) + (X-x_i)^{W} Q_i(X),$$

and dividing by $(X-x_i)^{W-w_i}$ this is $B = (X-x_i)^{w_i} Q_i + R_i$ with $\deg R_i < w_i$: division of B by $(X-x_i)^{w_i}$, whose remainder and quotient exist and are unique. The remainder is $B \bmod (X-x_i)^{w_i}$ by definition, the quotient is $(B-R_i)/(X-x_i)^{w_i}$, and $\deg Q_i = \deg B - w_i \le W-2$. Finally $R_i = 0$ iff $(X-x_i)^{w_i}$ divides B, since R_i is the remainder. □

Theorem 2. *The scheme of Fig. 1 satisfies Definition 4. Correctness holds even when registration labels coincide or τ equals a label.*

Proof. An honestly generated ciphertext carries a valid proof, so PartDec does not reject it. For every party, the honestly generated partial decryption verifies:

$$(\mathsf{sk}_i^{-1}\mathsf{ct}_2) \circ [\mathsf{sk}_i]_2 = \mathsf{ct}_2 \circ [1]_2.$$

Conversely, non-degeneracy of the pairing and $\mathsf{sk}_i \neq 0$ imply that this is the unique response that verifies, so it suffices to establish reconstruction for honest responses.

For an authorized set of parties S, the selector B has $B(0) = 1$, $\deg B \le w - t \le W-1$, and a root of multiplicity w_i at x_i for every $i \in [n] \setminus S$. Substituting $K_i = \kappa_i$ in Lemma 2, so that $A = \mathsf{SK}$, and adding $\gamma B = \gamma + \gamma(B-1)$, we have

$$(\mathsf{SK}(X)+\gamma)\,B(X) = \gamma + \sum_{i\in S} \kappa_i (X-x_i)^{W-w_i} R_i(X) + \underbrace{\sum_{i\in[n]} \kappa_i (X-x_i)^{W} Q_i(X) + \gamma\big(B(X)-1\big)}_{Q(X)},$$

where the first sum runs over S only because $R_i = 0$ for $i \in [n]\setminus S$ by the lemma. Each element of the aggregation equation is computable from the published elements:

- B from $\{[\alpha\tau^{-j}]_2\}_{j\in[0,W-1]}$, because $\deg B \le w-t \le W-1$, so the exponents $j-(w-t)$ of $\alpha\tau^{-(w-t)}B(\tau)$ lie in $[-(W-1),0]$.
- Q from $\{[\kappa_i(\tau-x_i)^j]_2\}_{j\in[W,2W-2]}$ of every party and $\{[\gamma\tau^j]_2\}_{j\in[1,W-1]}$. For the first part, expand Q_i by Lemma 1, and since $\deg Q_i \le W-2$:

  $$\kappa_i (X-x_i)^W Q_i(X) = \kappa_i (X-x_i)^W \sum_{j\in[0,W-2]} q_{i,j}(X-x_i)^j = \sum_{j\in[W,2W-2]} \kappa_i\, q_{i,j-W}\,(X-x_i)^j.$$

  For the second part, $B(X)-1$ has no constant term because $B(0)=1$, and degree at most $W-1$.
- H_i from $\{[\mathsf{sk}_i\kappa_i(\tau-x_i)^j]_2\}_{j\in[0,W-1]}$ and since $\deg R_i < w_i$, by Lemma 1:

  $$\begin{aligned}\mathsf{sk}_i\kappa_i (X-x_i)^{W-w_i} R_i(X) &= \mathsf{sk}_i\kappa_i (X-x_i)^{W-w_i} \sum_{j\in[0,w_i-1]} r_{i,j}(X-x_i)^j \\ &= \sum_{j\in[W-w_i,W-1]} \mathsf{sk}_i\kappa_i\, r_{i,j-(W-w_i)}\,(X-x_i)^j.\end{aligned}$$

Thus the aggregate algorithm can always compute the witness and recover the message. □

5.2 Security

We prove security in the asymmetric generic bilinear group model (Section 3), modelling the proof of knowledge π of Fig. 1 as a simulation-extractable proof.

Theorem 3. *Figure 1 is a CCA-secure weighted silent threshold encryption scheme (Definition 5) in the generic bilinear group model, when π is a simulation-extractable zero-knowledge proof of knowledge.*

Proof. We proceed through a sequence of games. Write Cor for the corrupt set, $\mathsf{Hon} = [n] \setminus \mathsf{Cor}$, and $S \subseteq \mathsf{Hon}$ for the set of honest parties whose partial decryptions of the challenge the adversary requests. Admissibility requires $\sum_{i \in \mathsf{Cor} \cup S} w_i < t$. Partial decryption queries on all other ciphertexts are unrestricted.

Game_0. This is the game of Definition 5. $\mathcal{A}$ outputs $(n, \boldsymbol{w}, \mathsf{Cor})$. The challenger samples $\alpha, \tau, \gamma \leftarrow \mathbb{F}^*$, gives $\mathcal{A}$ handles to the crs, samples $(x_i, \kappa_i, \mathsf{sk}_i) \leftarrow (\mathbb{F}^*)^3$ for $i \in \mathsf{Hon}$ and gives $\mathcal{A}$ handles to their public keys. $\mathcal{A}$ outputs randomness ρ_i for $i \in \mathsf{Cor}$, from which the challenger derives $(x_i, \kappa_i, \mathsf{sk}_i)$. The challenger computes $(\mathsf{ek}, \mathsf{dk})$ and gives them to $\mathcal{A}$. Partial decryption queries (i, ct) with $i \in \mathsf{Hon}$ are answered by $\mathsf{PartDec}(\mathsf{ek}, \mathsf{sk}_i, \mathsf{ct})$, before and after the challenge. $\mathcal{A}$ outputs $(t, \mathsf{msg}_0, \mathsf{msg}_1)$ with $\sum_{i \in \mathsf{Cor}} w_i < t \le w$. The challenger samples $b \leftarrow \{0,1\}$ and $r \leftarrow \mathbb{F}^*$ and returns handles to

$$\mathsf{ct}^* = \big(t,\ r \cdot \mathsf{com}_{\mathsf{pk},t},\ [r]_1,\ \mathsf{msg}_b + r \cdot [\gamma]_T,\ \pi^*\big).$$

Queries on ct^* itself are answered for $i \in S$ subject to admissibility. Finally $\mathcal{A}$ outputs b'.

Game_1*:* The challenger produces π^* using the simulator of the proof system instead of the witness r. By zero knowledge, the two games are computationally indistinguishable.

Game_2*:* For every partial decryption query (i, ct) with $\mathsf{ct} \neq \mathsf{ct}^*$ whose proof verifies, the challenger runs the extractor to obtain $r' \in \mathbb{F}$ with $\mathsf{ct}_2 = [r']_1$ and answers with $r' \cdot [\mathsf{sk}_i^{-1}]_1$, which equals the honest response. If extraction fails the game aborts, which happens with negligible probability by simulation extractability (Definition 2). Note that $\mathcal{A}$ may query $\mathsf{ct} \neq \mathsf{ct}^*$ with $\mathsf{ct}_2 = \mathsf{ct}_2^*$. Extraction then yields $r' = r$, which a generic adversary cannot produce except with negligible probability. Thus every extracted r' is a constant, and every non-challenge answer is a known multiple of $[\mathsf{sk}_i^{-1}]_1$.

Game_3*:* The elements $\mathsf{com}_{\mathsf{pk},t}$ and dk are deterministic linear combinations of the crs and the public keys, so $\mathcal{A}$ can compute them itself and we only give it $r \cdot \mathsf{com}_{\mathsf{pk},t}$. For $i \in \mathsf{Cor}$ the challenger gives $\mathcal{A}$ the scalars $(x_i, \kappa_i, \mathsf{sk}_i)$ instead of handles, which $\mathcal{A}$ knows anyway. Since every element removed from the view is computable from the remaining ones, $\Pr[\mathsf{Game}_3 = 1] = \Pr[\mathsf{Game}_2 = 1]$.

Game_4*:* The challenger replaces the payload by $\mathsf{msg}_b + [\rho]_T$ for an independent $\rho \leftarrow \mathbb{F}$. The view is now independent of b, so $\Pr[\mathsf{Game}_4 = 1] = 1/2$.

It remains to bound $|\Pr[\mathsf{Game}_3 = 1] - \Pr[\mathsf{Game}_4 = 1]|$, for which we apply the master theorem (Theorem 1). We work in the symbolic group model: each hidden scalar becomes a formal variable, and each handle in the adversary's view is labelled by a Laurent polynomial in these variables. The formal variables are

$$\boldsymbol{Y} = \big(X,\ A,\ \Gamma,\ E,\ (Z_i)_{i \in \mathsf{Hon}},\ (K_i)_{i \in \mathsf{Hon}}\big),$$

standing for τ, α, γ, r, the honest blindings sk_i, and the honest keys κ_i. Labels, weights, and the corrupt parties' secrets are constants known to $\mathcal{A}$. Let

$$\mathsf{SK}(X) = \sum_{i \in \mathsf{Cor}} \kappa_i \, (X - x_i)^{W - w_i} + \sum_{i \in \mathsf{Hon}} K_i \, (X - x_i)^{W - w_i}, \qquad F(\boldsymbol{Y}) = A^{-1} X^{w-t} \big(\mathsf{SK}(X) + \Gamma\big),$$

so that F labels $\mathsf{com}_{\mathsf{pk},t}$. The lists corresponding to the view in Game_3 are

$$L_1 = \Big(1,\ \{A^{-1}X^j\}_{j\in[0,2W-2]},\ \{\Gamma A^{-1}X^j\}_{j\in[0,W-1]},\ \{A^{-1}K_iX^j\}_{i,\,j\in[0,2W-2]},\ \{Z_i^{-1}\}_i,$$
$$\underbrace{E\cdot F,\ E,\ \{E\,Z_i^{-1}\}_{i\in S}}_{\text{challenge and responses}}\Big),$$
$$L_2 = \Big(1,\ \{AX^{-j}\}_{j\in[0,W-1]},\ \{X^j\}_{j\in[0,2W-2]},\ \{\Gamma X^j\}_{j\in[1,W-1]},\ \{Z_i\}_i,$$
$$\{Z_iK_i(X-x_i)^j\}_{i,\,j\in[0,W-1]},\ \{K_i(X-x_i)^j\}_{i,\,j\in[W,2W-2]}\Big),$$
$$L_T = (\Gamma), \qquad f = E\,\Gamma,$$

where i ranges over Hon when implicit. We now prove independence of f (Theorem 1) in Lemma 3, which completes the proof by showing that Game_3 and Game_4 are computationally indistinguishable. □

Lemma 3. *For every threshold t and admissible (Cor, S), i.e. $\sum_{i\in\mathsf{Cor}\cup S} w_i \le t-1$, if the honest labels are distinct then the polynomial $f = E\Gamma$ is independent of (L_1, L_2, L_T).*

Proof. Suppose towards a contradiction that $E\Gamma \in \mathrm{span}(\mathcal{C}(L))$. We compare coefficients of the independent variables one at a time.

Step 1: the variable E. Every element of L_2 and L_T has E-degree 0, and every element of L_1 has E-degree 0 or 1. In particular the elements Z_i^{-1} obtained from non-challenge decryption queries have E-degree 0 and play no further role. The E-linear part of a dependence is therefore a combination of products of the E-linear entries of L_1 with elements of L_2. Dividing by E, there exist polynomials $b, g, e_i \in \mathrm{span}\, L_2$ ($i \in S$) with

$$\Gamma = F\cdot b + g + \sum_{i\in S} Z_i^{-1}e_i. \tag{3}$$

Step 2: the variable A. Since A occurs in L_2 only in $\{AX^{-j}\}_{j\in[0,W-1]}$, every $h \in \mathrm{span}\, L_2$ decomposes uniquely as $h = A\,a_h(X) + c_h$ with

$$a_h \in \mathrm{span}\{X^{-j}\}_{j\in[0,W-1]}, \qquad c_h \in \mathrm{span}\big\{1,\ X^j,\ \Gamma X^k,\ Z_i,\ Z_iK_i(X-x_i)^j,\ K_i(X-x_i)^{j'}\big\}$$

with the index ranges as described above in L_2. We apply this to $h \in \{b, g, e_i\}$ from (3). The left side of (3) has A-degree 0. The only A^{-1} terms come from $F\cdot c_b$, so $c_b = 0$. The A^1 terms give $a_g + \sum_i Z_i^{-1}a_{e_i} = 0$, which we do not need. The A^0 terms give

$$\Gamma = \big(\mathsf{SK}(X) + \Gamma\big)P(X) + c_g + \sum_{i\in S} Z_i^{-1}c_{e_i}, \qquad P(X) := X^{w-t}a_b(X). \tag{4}$$

Since a_b involves only the exponents $-(W-1),\dots,0$, the Laurent polynomial P has exponents in $[w-t-W+1,\ w-t]$ and in particular none exceeds $w-t$.

Step 3: the variable Γ. Compare in (4) the terms that are linear in Γ and free of all K and Z variables. The left side contributes Γ. On the right, $\Gamma P(X)$ contributes, and c_g contributes $\Gamma c_g^{\Gamma}(X)$ with $c_g^{\Gamma} \in \mathrm{span}\{X^k\}_{k\in[1,W-1]}$. The Γ-terms of $Z_i^{-1}c_{e_i}$ carry a factor Z_i^{-1} and are different monomials. Hence

$$P(X) = 1 - c_g^{\Gamma}(X).$$

Thus P is a polynomial with $P(0) = 1$, and together with the exponent bound of Step 2, $\deg P \le w - t$.

Step 4: the variables K_i. For any $i \in \mathsf{Hon}\setminus S$, compare in (4) the terms that are linear in K_i and free of Γ and of all Z variables. The left side has none. On the right, $\mathsf{SK}(X)P(X)$ contributes $K_i\,(X-x_i)^{W-w_i}P(X)$, and c_g contributes $K_i\,s(X)$ for some $s \in \mathrm{span}\{(X-x_i)^j\}_{j\in[W,2W-2]}$, the unblinded hints of party i.

No other term on the right side qualifies. In c_g, the only remaining basis elements involving K_i are the blinded hints $Z_i K_i (X - x_i)^j$, which have Z-degree 1. It remains to consider the responses. For $s \in S$, the element e_s is what the adversary pairs with the response $E\, Z_s^{-1}$ of party s, and c_{e_s} is its A-free part, a combination of the same basis elements as c_g. Multiplying each of them by Z_s^{-1} yields

$$Z_s^{-1}, \quad Z_s^{-1} X^j, \quad Z_s^{-1} \Gamma X^k, \quad Z_s^{-1} Z_{i'}, \quad Z_s^{-1} Z_{i'} K_{i'} (X - x_{i'})^j, \quad Z_s^{-1} K_{i'} (X - x_{i'})^{j'},$$

with $i' \in \mathsf{Hon}$. The first four involve no K_i. The fifth involves K_i only for $i' = i$, and is then $Z_s^{-1} Z_i K_i (X - x_i)^j$, which is free of Z only if $s = i$ but this is impossible because $s \in S$ and $i \notin S$. The sixth has Z-degree -1. In other words, the response of party s can unblind only the remainder hints of party s itself, and no response of a party in S helps with the blinded hints of $i \notin S$. Therefore the coefficient of K_i must vanish:

$$(X - x_i)^{W - w_i} P(X) = -s(X) = (X - x_i)^W Q_i(X)$$

for some $Q_i \in \mathbb{F}[X]_{\leq W-2}$, because every element of $\operatorname{span}\{(X - x_i)^j\}_{j \in [W, 2W-2]}$ is $(X - x_i)^W$ times a polynomial of degree at most $W - 2$. Cancelling the factor $(X - x_i)^{W - w_i}$ gives $P(X) = (X - x_i)^{w_i} Q_i(X)$. In the terms of Lemma 2, the adversary has produced the decomposition of $(X - x_i)^{W - w_i} P$ with the remainder R_i of party i equal to zero, which is only possible if P has a root of multiplicity w_i at x_i. Since P does not depend on i, this holds simultaneously for every $i \in \mathsf{Hon} \setminus S$. Thus the polynomial P must have a root of multiplicity w_i at the label of every honest party that did not respond.

Step 5: counting. The honest non-responding parties have total weight $w - \sum_{i \in \mathsf{Cor} \cup S} w_i \geq w - t + 1$. Their labels are distinct, so the factors $(X - x_i)^{w_i}$ of Step 4 are pairwise coprime and their product, of degree at least $w - t + 1$, divides P. By Step 3, P is a nonzero polynomial of degree at most $w - t$, a contradiction. Hence $E\Gamma \notin \operatorname{span} \mathcal{C}(L)$. □

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